



About Citadel Securities

Citadel Securities is the next-generation capital markets firm and a leading global market maker. We provide institutional and retail investors with the liquidity they need to trade a broad array of equity and fixed income products in any market condition. The brightest minds in finance, science and technology use powerful, advanced analytics to solve the market's most critical challenges, turning big ideas into real-world outcomes.

Quantitative Researcher / Quantitative Research Analyst

Miami or NYC

The Quantitative Researcher at Citadel Securities conducts research and statistical analyses, develops automated trading strategies, and works with large datasets to predict market patterns. Responsibilities include creating and improving mathematical models and back testing trading signals in live environments.

Role Summary

At Citadel Securities, a leading global market maker, our team of quantitative researchers models the markets and brings trading strategies to life every day. Specifically, this team develops and tests automated quant trading strategies using sophisticated statistical techniques. We're looking for an extraordinary quantitative researcher who is committed to our core values that include winning, acting with integrity, continuously learning, and cultivating a meritocracy. Opportunities may be available from time to time in any location in which the business is based for suitable candidates.

Objectives

- Conduct research and statistical analyses in the evaluation of securities
- Work with large data sets, including unconventional data sources, to predict and test statistical market patterns
- Conceptualize valuation strategies, develop and continuously improve mathematical models, and translate algorithms into code
- Back test and implement trading models and signals in a live trading environment

Skills and Preferred Qualifications

- Advanced training in mathematics, statistics, physics, computer science, or another highly quantitative field
- Proficiency in probability & statistics (e.g. time-series analysis, machine learning, pattern recognition, NLP)
- Prior experience working in a data driven research environment
- Hands on programming experience in scripting (e.g. Python), analytical packages (e.g. R, Matlab) and/or compiled languages (e.g. C++)
- A background demonstrating strong analytical problem solving skills
- An ability to communicate advanced concepts in a concise and logical way
- Proficiency in creating and using algorithms to meticulously investigate and work through large data or error-checking problems

The employee who fills this role will be eligible to participate in a discretionary incentive compensation program, as well as a wide array of benefit programs, such as medical and life insurance, retirement and tax-free savings plans, and access to other healthcare programs.